AI Pathways Technical Track

Capstone Reflection Journal

As you progress through Course 3, this Capstone Reflection Journal offers a chance to pause, reflect on your learning, and take stock of the challenges and opportunities as you apply your new skills. Below you will find four prompts corresponding with the stages of the *Machine Learning Project Lifecycle*, plus a final, closing reflection.

Upon completion of the capstone, upload this journal in the Capstone Submission Portal, as per the instructions in the course.

SCOPING

Reflection Question: Now that you've reviewed the project plan given to you, how would you explain the 'big picture' goal in your own words? Does the project feel realistic to you? Is there a specific part that looks extra challenging?

The goal for me is provide the regulators and investors a tool to see which facilities are su

DATA AND MODELING

Reflection Question: As you experimented with the model, what settings or features would you consider changing to try to get better results? Did you discover anything messy, missing, or otherwise problematic about the data? If so, how did you fix it so your model could use the information? If you had to explain how this model makes its 'decisions' in simple terms, how would you describe it?

Adding features like co2 share would have help immensely but the whole idea was to find

DELIVERY

Reflection Question: How do you know your model actually works? Was it as accurate as you hoped, and if not, what is one thing you would change to make it better?

Model works because i used the cross-validation and confusion-matrix and the classificati

DEPLOYMENT

Reflection Question: Now that your project is ready for others to use, who do you think will benefit from it the most? What is one thing that might go wrong when a real person starts using the results of the model?

Govt and environmental agencies might benefit from it most , the model isn't perfect with

FINAL REFLECTION QUESTIONS FOR COURSE 3

1. You just built a model on real, messy data. Did it behave the way you expected? If not, what did you think was going to happen?

Not really ,Since most facilities were CCS candidates i hoped model would get highly ac

2. The class imbalance was left in on purpose. Did you notice it before you read this, or did your results tip you off? What was your first instinct when you saw the classification report?

I printed out the class distribution and saw 87% vs 13%. B When I saw the CU candidate

3. We picked the Decision Tree because people can follow its logic. But looking at your results, do you think that was the right call for this data, or did we leave performance on the table?

For this project yeah I think so. The whole point is being able to explain to someone why

4. Synthetic data, which is artificially generated to mimic real data and is often used to expand dataset size or balance classes, was an option we chose not to pursue. Knowing how the model performs, if you were designing this project from scratch, would you consider using synthetic data? What changes would you make to improve the model or the dataset?

No I would not use synthetic data , i would use that only if we didn't have any data to beg

5. Imagine handing this model to someone who has to decide where to spend the next hundred million dollars on carbon capture. What would you tell them it can do, and more importantly, what would you warn them it cannot?

I'd say this model can help you make a shortlist of facilities to look into for carbon captur

6. What is one thing you would fix if you worked with this data again?

I would add more features. Three features (emission scale, industry sector, reporting his

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